Dated: 22/11/22 Make Up Sessional Test Max Marks: 15

Automata Theory

B.Tech Comp 5th Sem, JMI

Q1) Show that Show that **(0+1)*01(0+1)*+1*0* = (0+1)*** **(2)**

Q2) Verify the following statement: "***Subset of every regular language is not regular***". **(3)**

Q3) Convert the following CFG to CNF along with mentioning appropriate rules wherever needed. **(5)**

S → AbSCd | aAc

B → BA| a | ε

A → BC | ε

C → c

Q4) Design and verify a PDA for a language L = $\{a^n c^{n+1}, n>=0\}$ **(5)**

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